COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 010- DATA STRUCTURE AND ALGORITHMS	**SECTION:**	CPE21S4
DATE PERFORMED:	OCT. 9, 2026	**DATE SUBMITTED:**	OCT. 12, 2026
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ACTIVITY NO. 11.1 PLANNING

1. PROCEDURE OUTPUT

5 PROPOSED SYSTEM

Airbnb Damage and Missing item Monitoring System

Once guests have checked out, hosts must assess if any furniture, appliances, kitchen utensils, towels, or other amenities are absent or damaged. This system assists Airbnb hosts in tracking the state and availability of their property belongings both prior to and following each guest's visit. It enables hosts to log current items, note damages, recognize absent belongings, and refresh item statuses via an easy tracking method. It can also keep logs of past inspections, monitor reported incidents, and create summaries of lost or damaged items

10 QUESTIONS

1. How do you currently list and organize the items inside your Airbnb property?
2. How do you record the quantity and condition of each item before a guest checks in?
3. How do you check and update item records after a guest checks out?
4. How do you identify items that are missing or damaged?
5. How do you search for a specific item in your property inventory?
6. How do you compare the item records before and after a guest's stay?
7. How do you organize records of damaged and missing items for easy access?
8. How do you calculate the total estimated cost of damaged or missing items?
9. How do you sort or prioritize damaged items based on their condition or repair cost?
10. What problems do you experience when managing item records manually?

Print Shop Queue Tracker

In a print shop, customers drop off files for documents, tarpaulins, or photos, and the staff has to figure out what to print first and remember who is waiting for what. When the shop gets busy, jobs pile up, and it becomes easy to lose track of an order or to print an urgent job too late. The Print Shop Job Queue and Tracking System keeps every order in one place. It lines up the jobs by urgency and deadline, lets the staff pull up any order using its ticket number, and shows whether a job is waiting, printing, done, or cancelled. At the end of the day, it can also list how many jobs were finished and which ones are still pending

10 QUESTIONS

1. Who are the people involved in handling a print order, from the counter to the printer, and what does each person do?
2. What happens first when a customer brings in a file or document to be printed?
3. What information about the order do you collect and write down at that point?
4. After an order is accepted, what steps does it go through before it is ready for pickup?
5. How do you decide the order in which the waiting jobs are printed?
6. What changes in the process when an order is a rush order or has a very short deadline?
7. How do you keep track of where each order currently is, and how do you find it when a customer asks about it?
8. What happens when a job has a problem, such as a printing error, a missing file, or a customer who cancels or changes the order?
9. How do you hand over the finished order, and how do you record that it was paid for and picked up?
10. What records or reports do you keep from these orders, and what do you use them for?

Event Planner Task Dependency System

Organizing an event such as a wedding, a debut, or a school program involves many small tasks, and a lot of them cannot start until another task is finished. For example, invitations cannot be sent before the venue is booked, and the caterer cannot be confirmed before the number of guests is known. The Event Planner Task Dependency System helps an organizer put all of these tasks in one place, link each task to the ones that must come before it, and arrange them automatically into an order that makes sense. The organizer can also assign each task to a person, set its due date, and mark it as not started, ongoing, or done. When something is delayed, the system shows which other tasks are affected. After the event, the finished task list is saved so it can be reused as a starting point for the next event.

10 QUESTIONS

1. What kinds of events do you organize, and who is involved in preparing one, from the client to the suppliers?
2. What is the first thing you do when a client books you for an event?
3. How do you list down all the tasks that need to be done before the event day?
4. How do you know which tasks must be finished before another one can start, such as booking the venue

before sending the invitations?
5. How do you decide who is responsible for each task and when it should be done?
6. How do you keep track of which tasks are done, ongoing, or not yet started?
7. What do you do when a task is delayed or a supplier backs out, and how does that affect the other tasks?
8. How do you coordinate with suppliers such as the venue, caterer, and photographer, and how do you record what each one agreed to provide?
9. What do you do in the last few days before the event to make sure nothing is forgotten?
10. After the event, what records do you keep, and how do you use them when planning the next one?

Badminton Court Booking System

Badminton courts are usually rented by the hour, and the owner has to make sure that two players never end up with the same court at the same time. In many places, bookings are still written in a notebook or sent through chat, so double bookings and mix-ups happen easily, especially on busy evenings and weekends. The Badminton Court Booking System lets the owner record each reservation with the court, the player's name, and the start and end time. Before a booking is accepted, the system checks whether the time overlaps with an existing one and suggests the next free time if it does. The owner can also cancel or move a booking, view each court's schedule for the day, and keep a record of past bookings and the total hours rented.

10 QUESTIONS

1. How many courts do you have, and what are your opening and closing hours?
2. How do players usually reserve a court, and who takes the reservation?
3. What details do you write down for each booking?
4. How do you check that a time slot is still free before you accept a booking?
5. What happens when two players ask for the same court at the same time?
6. How do you handle walk-in players when the courts are already reserved?
7. What do you do when a player cancels, arrives late, or asks to move their schedule?
8. How do you compute the fee for the time rented, and when and how do players pay?
9. How do you keep track of which courts are in use right now and which one becomes free next?
10. What records or reports do you keep from the bookings, and what do you use them for?

Parish Mass Intention Schedule

In a parish, people go to the church office to request a mass for a loved one, whether for a birthday, a thanksgiving, or someone who has passed away, and each mass can only take a limited number of intentions. The parish staff must keep track of which dates and times are already full, who is waiting for a free slot, and which intentions will be read at each mass. The Parish Mass Intention Scheduler lets the office record each request with the name of the person being prayed for, the kind of intention, the preferred date, and the offering. It checks whether the chosen mass still has room, and if it does not, it places the request on a waiting list or suggests the next available mass. The staff can also move or cancel a request, print the list of intentions for each mass, and keep a record of past requests.

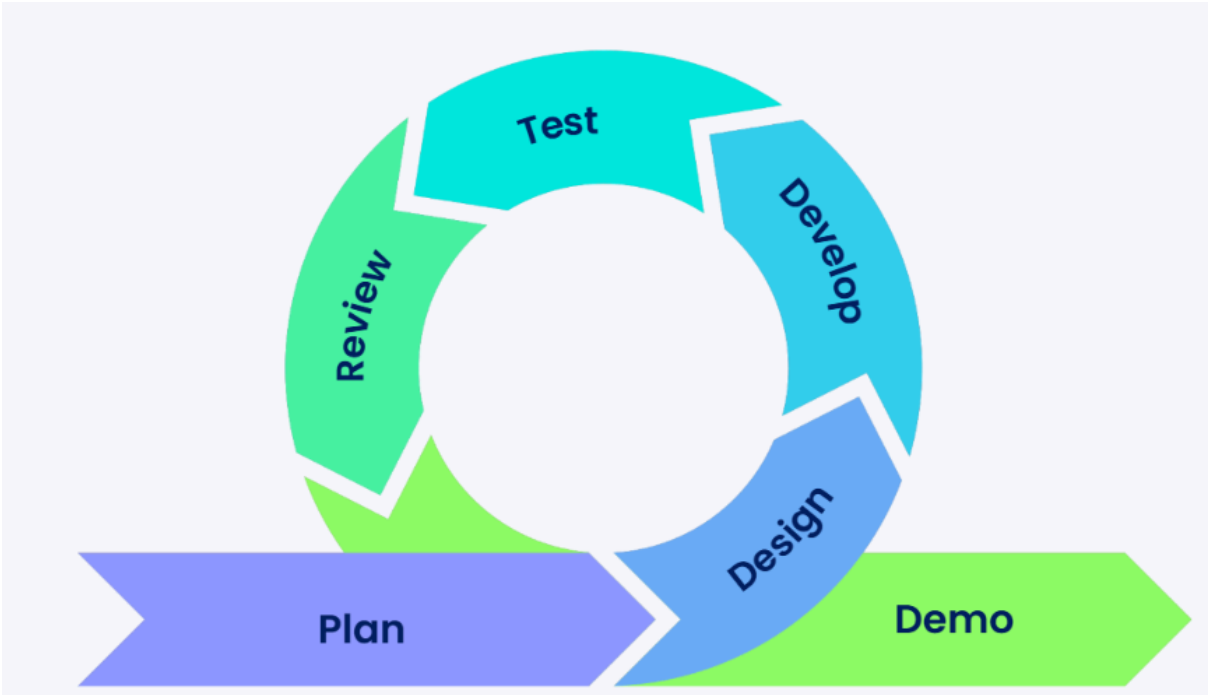
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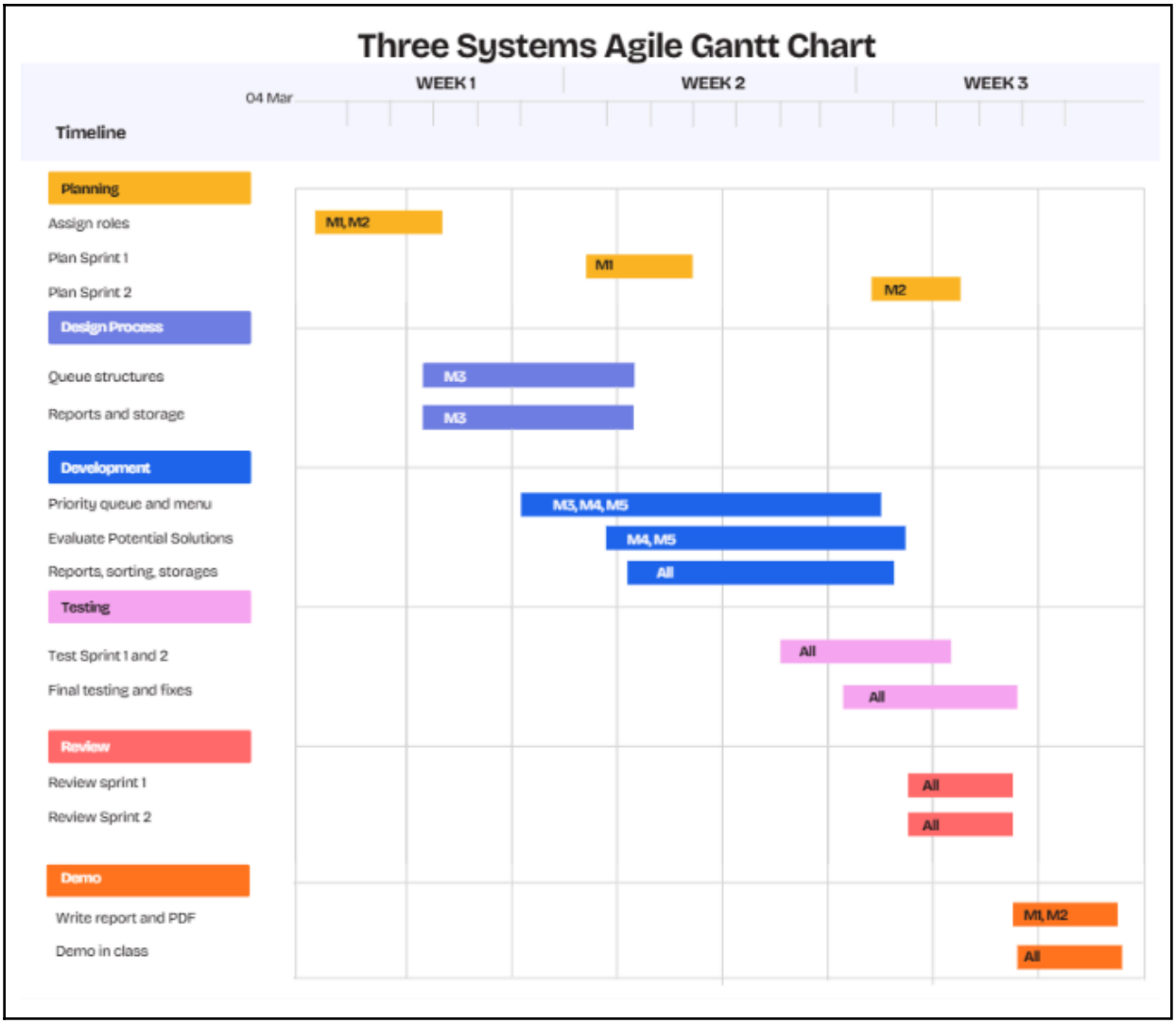
10 QUESTIONS

1. What are the regular mass schedules in your parish, and how many intentions can each mass take?
2. Who accepts the mass intention requests, and how do people usually make them (in person, by phone, or by message)?
3. What details do you write down for each request?
4. What kinds of intentions do you receive, and does the kind of intention affect when it is scheduled?
5. How do you check whether a mass already has too many intentions on a certain date?
6. What do you do when the date a person wants is already full?
7. How do you handle requests that are cancelled, moved to another date, or repeated for the same person?
8. How do you record the offering, and what do you give the person as proof of the request?
9. How do you give the priest or the commentator the list of intentions to be read at each mass?
10. What records of past intentions do you keep, and what do you use them for?

SDLC METHOD (AGILE)



GANTT CHART



2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

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I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: _____

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: _____

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: _____

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

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